Base-point and coefficient investigation — 29 October 2026

The historical skip criterion remains unknown. The requested wider comparison is prepared and tested on synthetic inputs, but has **not run on the actual 21-curve cohort**. The restored environment has no configured checkpoint-upload identity; AGENTS.md requires a successful uploaded checkpoint before an experiment starts.

What the G–b connection establishes

For the user's canonical point, the defining equation is $y^2 = x^3 - 3x + b \pmod p$. Thus $x=0$ implies $y^2=b$, and $x=1$ implies $y^2=b-2$. The supplied P-224 index 777 and P-256 index 876 points use $x=0$; P-224 indices 1485 and 5736 use $x=1$. A close numerical or algebraic relation between these assigned points and b is therefore expected. These points were chosen by the user's rule; they are not recovered historical generators.

The existing, independently checked complete-factor feature tables record b:chi_p=1 for every one of the 17 non-P-224 rows. Including P-224 index 777, b is square for nine of ten published and nine of eleven skipped curves. P-224 indices 1485, 5736 and published 7026 have character −1. The literal old measurements and member hashes are in evidence/existing-b-character.json and the inherited-feature custody record. This reuses prior results; it is not a new cohort execution.

Under the reference convention in the non-P-224 streams, $p\equiv 7 \bmod 8$ and $b=R^{(p+1)/4} \bmod p$. The exponent is even, so this b is square. The first-x point rule consequently starts at $x=0$ for both labels. Conditional on the seed-generation equation $c b^2=-27 \bmod p$, $x=0$ also implies $c y^4=-27$; $x=1$ implies $c(y^2+2)^2=-27$. These are consequences of the stated equations, not a selection mechanism or an attack. Five conditional Lean theorems compile; their transitive axiom audit passes using only standard foundations. The proof map states their assumptions and limits.

Model and generator controls

The historical P-224 coefficients for 777 and 5736 have the opposite sign to the user's smaller-b representatives. Keep these models distinct. Since P-224 $p\equiv 1 \bmod 4$, choosing $i^2=-1$ transports b to −b by $(x,y)\to(-x,iy)$. Substitution negates both sides of the curve equation; the inverse map exists because $i\neq 0$. This readable derivation and native toy controls are not a formal proof of the full elliptic-curve isomorphism. The prepared real-curve checks must verify the actual transported generators. Do not identify opposite signs as same-order models over $p\equiv 3 \bmod 4$.

The comparison retains canonical points, ten standardized published generators and **eleven unknown historical skipped generators** separately. It tests both signs of seven canonical-point multiples and 22 public scalar recipes. A known-scalar equality test is not discrete-log recovery. A relation found only for an assigned canonical point cannot explain how an unavailable historical generator was chosen. Implementation and protocol preconditions would still be needed to turn any trait into a vulnerability.

Prepared scope and actual checks

There are infinitely many functions of curve parameters. The finite inventory is explicit: the existing 1,127 traits, plus 171 point/coefficient traits across two coefficient views and 14 point choices. The views and point choices are controls on the same 21 curves, not additional independent historical samples. Planned outputs include all 25 same-field published/skipped pairs, 28,175 inherited differences, 119,700 point-trait differences and 19,152 tight published-envelope comparisons. No resulting actual-curve values or matches are claimed. All labels are already visible; any fitted rule is exploratory.

The actual development check uses six tiny prime-field examples repeated into a 21-row parser fixture, with planted G=[b]Q relations and dummy inherited features. The separate GMP verifier checked its 100,548 trait cells, 19,152 envelopes, 119,700 point differences and 28,175 dummy-feature differences. Nine deliberately corrupted outputs were rejected for the intended reasons. The fixture's 106 recipe matches and zero exact single separators are **toy results only**, with no implication for the NIST cohort. See the development report, raw-bundle custody and control receipts.

Production flow: one comparison planned; zero attempted; zero completed; zero curves excluded. No DLP, timing benchmark or paid experiment ran.

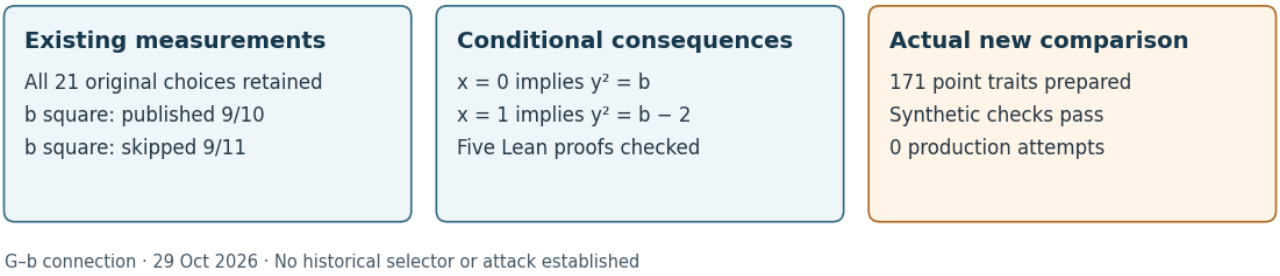
Requirements and remaining obligations

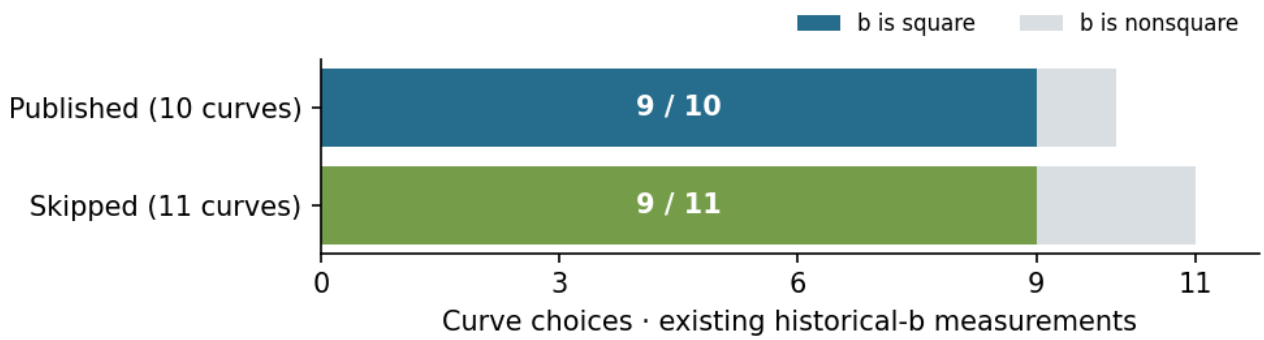
Request	Status and evidence
Original four ICV1/DiSSECT assessments	Previously delivered in the original study.
Four remaining complete factorizations	Verified complete: 84/84 records across all 21 curves. Unknowns were not numeric labels or silently dropped.
Comprehensive differences and G–b investigation	Partial: equation consequences and existing traits assessed; finite native comparison implemented and synthetically checked; actual new measurements blocked.
Historical criterion explaining skips	Unresolved. Existing post-hoc separators and failed fixed hypotheses do not establish original use. The user has no original visit/acceptance logs.
Weak selected or weak skipped curves	Neither established. No applicable attack construction or measured DLP.
Updated PDF and PR	This report/PDF and existing PR 2666; preparation delivery and actual cohort comparison remain open.

Restore the existing checkpoint identity through runtime settings, admit the preparation bundle, freeze the supported execution route and run the full comparison with independent verification. Keep failures, zero recipes and unknown historical generators explicit. A separation would justify a stated prospective test; a negative comparison would exclude only this finite family. Neither outcome alone supplies historical intent or a security verdict.

Sources: SEC1v2 §3.1.3.2, RFC5639 A.2, pinned std-curves, prior source check, PROTOCOL.md, REPRODUCE.md. SEC1 and RFC5639 supply methods to compare; they do not authenticate the 1997 caller or historical skip rule. Standard-source characteristics are not substituted for independently measured traits.

The new diagram describes this evidence boundary. Existing factor, embedding, FreeLIP and ECPP-work figures remain unchanged because no new actual-cohort numerical result supersedes them. No leaderboard, performance gain or security metric is promoted. Storage migrations in upstream changed some old metadata bindings; additive custody observations preserve those differences without rewriting frozen manifests.




b is square
b is nonsquare
Published (10 curves)
Skipped (11 curves)
9 / 10
9 / 11
0
3
6
9
11
Curve choices · existing historical-b measurements